

LAB-03 Data Visualization

Scenario

You are hired as a Business Intelligence Intern at **PakTech Electronics Pvt Ltd**. The company sells electronics and accessories across Pakistan in multiple regions. Management wants you to build a **Power BI dashboard** and perform data analysis using DAX. You are provided raw sales data in CSV format.

Tasks:

- Import the data into Power BI
- Create calculated columns
- Create measures
- Perform analysis using DAX
- Build an interactive dashboard

You are NOT allowed to copy-paste formulas. All DAX must be written manually.

Row Level Calculations (Calculated Columns)

These are computed for **each individual row (order line)** using row context.

1 Gross Sales

Business Meaning: Total selling amount before discount.

Formula: $\text{Gross Sales} = \text{Quantity} \times \text{Unit Price}$

Example: Quantity = 2, Unit Price = 700 → Gross Sales = 1400

2 Discount Amount

Business Meaning: Amount reduced from gross sale.

Formula: $\text{Discount Amount} = \text{Gross Sales} \times (\text{Discount \%} / 100)$

Example: $1400 \times 10\% = 140$

3 Net Sales

Net Sales = Gross Sales – Discount Amount

4 Total Cost

Total Cost = Quantity × Unit Cost

5 Profit

Profit = Net Sales – Total Cost

6 Profit Margin (Row Level)

Profit Margin = (Profit / Net Sales) × 100 (0 if Net Sales = 0)

Aggregated Measures (Filter Context)

1 Total Net Sales

Total Net Sales = Sum of Net Sales (filtered rows)

2 Total Profit

Total Profit = Sum of Profit

3 Total Quantity Sold

Total Quantity = Sum of Quantity

4 Total Orders

Total Orders = Count of OrderID

5 Unique Customers

Unique Customers = Count of distinct CustomerID

6 Average Order Value

Average Order Value = Total Net Sales / Total Orders

7 Maximum & Minimum Sale

Max Sale = Maximum Net Sales, Min Sale = Minimum Net Sales

Logical Business Measures

1 Profit Status Classification

Profit > 500 → High, 200–500 → Medium, < 200 → Low

2 Sales Performance Rating

Total Net Sales > Target → Excellent, else Normal/Poor

3 Region Code Mapping

North = 1, South = 2, East = 3, West = 4

Overall Profit Margin

Overall Profit Margin = (Total Profit / Total Net Sales) × 100 (0 if denominator = 0)

Percentage of Total (ALL Concept)

Region % of Total = (Region Sales / Grand Total Sales) × 100

Filter-Based Calculations

- North Region → Net Sales where Region = North
 - Electronics Category → Net Sales where Category = Electronics
 - Quantity > 2 → Net Sales where Quantity > 2
 - Female Customers → Net Sales where Gender = Female
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Advanced Conditional Filtering

- Profit > 300 → Net Sales sum
 - Discount > 10% → Net Sales sum
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Context Understanding

- **Row Context:** Calculation per row, unaffected by slicers
 - **Filter Context:** Aggregation changes with applied filters
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Cost Percentage

Cost % = (Total Cost / Total Net Sales) × 100

Discount Impact Analysis

- Total Discount Given = Sum of Discount Amount
 - Discount % of Sales = (Total Discount / Gross Sales) × 100
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Key Technical Concepts

1. Aggregation = Sum over filtered dataset
2. Conditional logic = Rule-based classification
3. Percentage = $\text{Part} / \text{Whole} \times 100$
4. Filtering = Subset selection before aggregation
5. Context = Scope of calculation (Row vs Filter)
6. All DAX must be written manually without copy-paste